

# Open Source IoT: A Primer for Everyone

Open source IoT platforms promise to play a central role in shaping the future, making it possible for more people and businesses to benefit from smarter, more efficient solutions. We look at a brief history of this tech and explore emerging trends.



**V**isualise waking up in a home where the lights gently brighten as your alarm rings. The AC adjusts to your preferred temperature, and your car sends you a message that it's fully charged and ready for the day. All of this happens automatically, without you lifting a finger, because the devices in your home are communicating with each other. This is the magic of the Internet of Things (IoT), where everyday objects are connected to the internet, allowing them to work together in an intelligent, automated way.

Now, imagine that the software and tools powering this interaction are

not locked behind expensive licences or proprietary systems. Instead, they are open source—free to use, modify, and share. This allows developers everywhere to come up with new ideas and solutions. For example, many people use proprietary smart home systems like Google Nest for controlling thermostats or Amazon Alexa for voice-based automation. While these systems are user-friendly, they can be expensive, and users are limited to the products and services offered by the prominent manufacturers.

In contrast, open source platforms like Tasmota allow ordinary devices, such as light switches or bulbs, to

become smart devices. Tasmota firmware can be installed onto compatible microcontrollers, which are small electronic components used to control devices. Once the firmware is installed, these devices can connect to Wi-Fi and be controlled through mobile apps. Unlike systems like Philips Hue or Samsung SmartThings, Tasmota doesn't require expensive, proprietary hardware. It repurposes existing devices, saving money while creating a customised smart home. Tasmota is also flexible, as it works with various devices and platforms, making it an ideal choice for those who want to experiment with technology or build smart systems on a small budget.